

In-the-Wild Exploration of the Impact of Sleep on Working Memory with a Wearable

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ABSTRACT

Working memory, essential for temporarily retaining information, is a significant indicator of performance in cognitive tasks, offering valuable insights into mental tasks such as the ability to learn. While prior lab experiments have shown that sleep affects working memory, it remains uncertain whether this is detectable in naturalistic settings. Due to its nature, assessing working memory entails complex processes that rely on short-term memory and information-processing abilities, posing significant challenges to assessors. Identifying alternative factors affecting working memory could allow for the estimation of working memory status via physiological signals. Our study explores whether physiological data obtained during sleep can predict working memory during real-world tasks. Findings from experiments involving 12 university students suggest that average heart rate and heart rate variability during sleep are indicators of working memory in such settings.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI; Empirical studies in ubiquitous and mobile computing.

KEYWORDS

working memory, in-the-wild, wearables, physiological sensing, cohort analysis

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1 INTRODUCTION

Working memory is the ability to temporarily hold information while performing cognitive tasks. Individuals with higher working memory capacities show better reading comprehension, arithmetic skills, and vocabulary learning effectiveness [3, 8, 20]. Thus, working memory serves as an important indicator that can predict performance in a wide range of cognitive tasks. Measuring working memory allows us to evaluate an individual's capacity for different forms of learning [25]. Working memory is not static and can depend on the physical and mental states of a person [4], thus making in-situ classification of working memory both possible and desirable.

Assessing working memory requires complex processing that combines short-term memory ability and information processing capability, which places a significant burden on the subjects [7]. Additionally, the measurement process takes significant time. One way to solve these problems is by (1) finding less obtrusive and demanding ways to assess working memory capabilities, e.g., by identifying factors in daily activities that influence working memory, and (2) then predicting working memory based on these factors. Kuriyama et al. [15] found that participants' post-sleep working memory improved significantly compared to pre-sleep. However, since this study was conducted under strict sleep schedule management, it is uncertain whether the results are applicable in naturalistic settings.

In our work, we aim to clarify the relationship between sleep and working memory in-the-wild by measuring sleep duration, duration of sleep phases and physiological measurements overnight in naturalistic settings and extracting relationships from the resulting data. We use the Oura Ring¹, which passively collects sleep data when worn. This minimizes the burden on the participants and provides accurate data. Different studies have confirmed the high accuracy and reliability of the data collected by the Oura Ring [5, 16].

We conducted an experiment with 12 Japanese undergraduate and graduate students over two weeks. We analyze sleep data collected from the Oura Ring and assess the participants' working memory every morning. Using linear mixed models for analysis, this study revealed a positive correlation between higher average heart rate (HR) and heart rate variability (HRV) during sleep and better working memory performance. These findings suggest that physiological data obtained from wearables like the Oura Ring can

¹<https://ouraring.com/>

serve as a means to estimate working memory capabilities outside controlled environments while simultaneously lowering the burden on the participant to a minimum. Our work stresses the significance of incorporating real-world data into cognitive research, offering valuable insights for both academic and practical applications.

2 RELATED WORK

For researchers to gauge the potential of real-world interventions, it is important to determine the correlation between physiological sensing data and various human outputs. Hence, selecting the most suitable task for measuring output becomes crucial. Given the diversity of cognitive states, using the correct types of tasks is essential for inferring these states accurately. For example, Tag et al. [23] uses a psychomotor vigilance task (PVT) to measure cognitive readiness, or alertness, in-the-wild, while Vargo et al. [24] uses an arithmetic exercise to measure the relationship between sleep and problem-solving ability. In our case, looking at working memory capabilities, the N-back test has been shown to be an appropriate tool [19]. In the N-back task, participants are shown a series of items, e.g., letters or shapes, and after every new item, they are tasked to recall the item shown in steps earlier. Typically, studies vary n across experimental blocks to evaluate how different memory demands affect behavioral performance and physiological responses [13].

Kato et al. [11] used the N-back test to investigate the relationship between sleep and working memory. In this study, they measured total sleep time, sleep efficiency, bedtime, and wake-up time using actigraphy over five to seven days in 23 healthy young adults (13 males, 10 females, aged 22.0 ± 2.2 years). They assessed their working memory by measuring the accuracy and reaction time on the N-back task with N values of 0, 1, and 2. This study showed a significant positive correlation between total sleep time and the accuracy of the N-back task with an N value of 2 ($r = 0.486, p = 0.019$). No significant correlation was observed between total sleep time and the accuracy of the N-back task with N values of 0 and 1, suggesting that adjusting the difficulty is important when investigating the relationship between sleep and working memory.

One significant limitation of Kato’s work is that this study was conducted under controlled conditions with physician confirmation of participants’ mental and physical health and specific sleep conditions. Thus, the applicability of the results to real-world environments remains unverified. In our study, we measure sleep and test working memory in an in-the-wild setting.

Additionally, in our analysis, we include more detailed sleep data, such as average HR during sleep and average HRV, a measure of the change in time intervals between two heartbeats, measured in milliseconds [17]. This allows us to provide a more comprehensive investigation of the relationship between sleep and working memory.

3 EXPERIMENTAL PROCEDURES

3.1 Participants

In this study, we used sleep and working memory data, obtained with an N-back test, from 12 students (11 male and 1 female; mean age, 21.9 ± 1.51 years [SD]) enrolled at a Japanese public university. Upon obtaining informed consent for research use of their data,



Figure 1: Oura Ring

participants were provided with an appropriately sized Oura Ring. Subsequently, participants continued their daily routines while their sleep was monitored and no specific directions were given regarding sleep or device usage. All participants were instructed to conduct working memory measurements as soon as possible after waking up in the morning. The measurements continued for two weeks (14 days) for all of the participants. Ethics approval was granted for this study by the ethics committee of the university where the study was conducted.

3.2 Sleep Measurement

Sleep measurements in this study were obtained using the Oura Ring, as shown in (Figure 1). The Oura Ring is a wearable sensing device in the form of a finger ring. It is equipped with an infrared optical HR sensor, a body temperature sensor, and a 3D accelerometer. The Oura Ring is a consumer device that can record bedtime and wake-up time, and it can measure sleep duration for each of the four sleep stages (*Awake*, *REM Sleep*, *Light Sleep*, *Deep Sleep*). Additionally, it can capture other data during sleep, such as body surface temperature, average HR, average breathing rate, latency (time to fall asleep), and average HRV. Body surface temperature represents the changes observed by comparing the baseline formed over a 2 month period of wearing the Oura Ring with the body surface temperature recorded in the most recent 3 days, using a weighted average. The average HR and average respiratory rate represent the mean values per minute, while latency indicates the time it takes to fall asleep. Average HRV refers to the millisecond-level variation between heartbeats and represents the average of all samples measured at 5-minute intervals throughout the sleep period.

3.3 Working Memory Measurement

In this study, we custom developed an N-back task mobile application to measure working memory. To control for factors other than sleep, participants were prohibited from eating, drinking alcohol, smoking, consuming caffeinated beverages, engaging in vigorous exercise (such as running), and taking showers or baths from the time they woke up until the N-back task was completed. Additionally, we introduced a normalized scoring system to evaluate working memory performance. The following sections will explain the N-back task, the N-back task mobile application, and the scoring system.

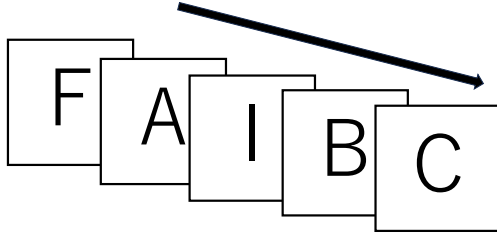


Figure 2: N-back task

The N-back task is a continuous performance task used to measure working memory, requiring participants to recall the stimulus presented N positions back in a sequence of continuously presented stimuli. Below, we will explain the N-back task with $N=3$, as used in this study, using Figure 2 as an example.

Looking at Figure 2, we assume that the letters are displayed sequentially from the back (F) to the front (C). First, the letters F, A, and I are displayed sequentially, and the participant must remember them in order. When B is displayed next, the participant must remember B and simultaneously recall the letter presented three positions back, which was F.

Next, when C is displayed, the participant must recall the letter presented three positions back, which was A. In this manner, after the initial three letters are displayed, the participant must recall each letter presented in three positions earlier each time a new letter is displayed. The N-back task involves repeating this operation for a specified number of trials.

The N-back task is conducted using a mobile application specifically created for this study. In this application, participants are presented with letters of the alphabet on the memory screen (as shown in Figure 3) for 1.5 seconds each, which they must remember. They then proceed to the response screen to answer which letter was presented three positions back. There is a series of a total of 20 letters in the task, and the letters displayed are randomly selected from A to I, i.e., from nine different options.

The data logged by this application include the timestamp of each measurement, the number of correct answers out of the series of 20 letters, and the time taken to answer.

To evaluate the working memory using the N-back task requires calculating accuracy rates and response times [9, 10, 21]. A high accuracy rate and short response time are indicative of high working memory performance and should thus be considered equally during evaluation. Therefore, in this study, we use a normalized score S expressed by the following equation using accuracy rate C , average response time T , maximum value of T ($T_{\max}$), and minimum value of T ($T_{\min}$).

$$S = C + \frac{T_{\max} - T}{T_{\max} - T_{\min}} \quad (1)$$

The second term on the right-hand side of Equation 1 involves normalization, and it is scaled to match the first term (accuracy rate C). Consequently, when the accuracy rate is high and the average response time is short, the adjusted score increases. Therefore, when the normalized score is high, it indicates high working memory.

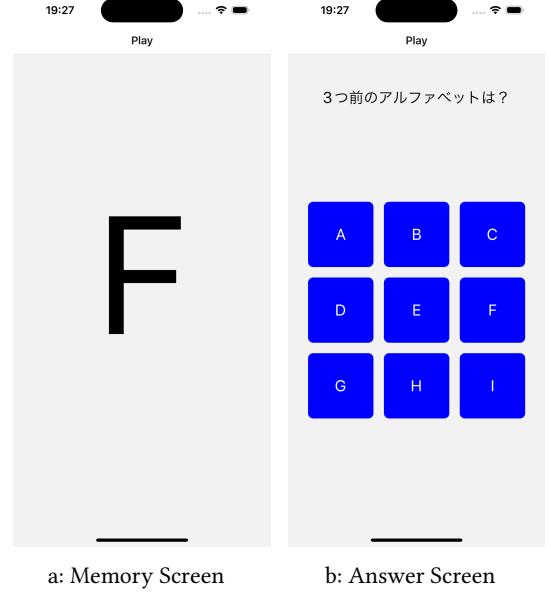


Figure 3: N-back Task Application

Table 1: 9 types of sleep data

Variable	Sleep Data
x_1	Awake Duration
x_2	Rem Sleep Duration
x_3	Light Sleep Duration
x_4	Deep Sleep Duration
x_5	Average HR
x_6	Average Breath Rate
x_7	Average HRV
x_8	Efficiency
x_9	Latency

4 RESULTS

Before the start of the experiment, participants were asked to perform one measurement with the N-back task application to familiarize themselves with the operation. However, it cannot be assumed that all participants became accustomed to the operation after just one measurement. Therefore, after conducting a post-experiment survey regarding the duration it took for participants to become familiar with the N-back task application, data from a total of 29 days were excluded based on each participant's responses, leaving us with data from 139 days. Additionally, one participant provided an ambiguous response indicating that they "[...] don't specifically remember", but they got accustomed to the application "[...] around the 3rd or 4th day." For analysis purposes, it was assumed that the participant became accustomed to the application by the 4th day, and data from before the 3rd day were not included in the analysis.

4.1 Model Selection

To analyze the obtained data, a linear mixed-effects model is employed with sleep data as the explanatory variables and adjustment scores as the response variables.

Table 2: Results of model₆

Variable	Estimated value	p-value
(Intercept)	5.908×10^{-1}	0.08659
Rem Sleep Duration	1.752×10^{-5}	0.08281
Average HR	1.076×10^{-2}	0.03000 *
Average HRV	5.591×10^{-3}	0.00116 **
Latency	6.419×10^{-5}	0.06347

* $p < 0.05$ ** $p < 0.01$

Among the nine types of sleep data obtained from the Oura Ring, as shown in Table 1, there is a possibility that some data may decrease the accuracy of the model. We calculate the AIC (Akaike Information Criterion) [1] for the 10 constructed models and performed model selection based on the AIC results. Multicollinearity was checked with Variance Inflation Factor (VIF) [2]. This allowed us to identify the best fitting model as shown in Table 2.

4.2 Analysis Result

The results of the analysis by model₆ are presented in Table 2. In Table 2, it can be seen that the adjusted score significantly increased as the values of average HR and average HRV increased. Higher accuracy and shorter average response time led to a higher adjusted score. Therefore, we infer that higher values of average HR and average HRV indicate higher working memory.

We assessed the goodness of fit of model₆ by calculating Nakagawa's R^2 [18]. The results clearly show that Conditional R^2 is higher than Marginal R^2 , indicating that the goodness of fit improved by adding random effects to the model to account for individual differences. Both results, while failing to explain the majority of the performance, are surprisingly high given the in-the-wild nature of the experiment and consequentially noisy data.

5 DISCUSSION AND LIMITATIONS

The overall results indicate that multimodal physiological data obtained via a consumer device like the Oura Ring can be used to partially predict performance on a working memory task. In addition, the fact that the task was done repeatedly in an in-the-wild setting lends significance to the analysis, clearly indicating its feasibility for real-world applications. While the results do not ascend to the level of being prescriptive, the results do support and inform the design of future efforts to do larger and longer studies.

There are some notable limitations and opportunities for future work. While the results show a significant increase in adjusted scores as both average HR and average HRV values increased, it is important to interpret these with caution. Our study involves a two-week research period with 12 students, resulting in a relatively small dataset. Additionally, the participants were all students from a Japanese public university, limiting the diversity among participants. Moreover, there was a gender imbalance, with 11 male participants and only 1 female participant. Future studies should have more inclusive participant pools accounting for cultural, age, and gender diversity to make the result generalizable.

Table 3: Nakagawa's R^2

Conditional R^2	0.3192
Marginal R^2	0.1918

Next, HRV is an often-used and validated indicator of stress, with reports [6, 12, 22] indicating that higher stress levels are associated with lower HRV. Therefore, it appears that sleep in a relaxed state might enhance working memory, which aligns with intuitive interpretations. However, it is also reported that HR increases with rising stress [14]. Consequently, analyzing the impact on working memory should consider both the increase in average HRV and the decrease in average HR due to relaxation. Thus, as future work, we plan to investigate the effects of stress on sleep in real-world environments.

Finally, the adjusted scores in this study did not account for the differences in the means and variances of the accuracy rate and average response time. Therefore, it might be necessary to consider these differences in variability and apply appropriate weighting to each when interpreting adjusted scores.

6 CONCLUSION

In this experiment, as a first step in constructing a model to estimate working memory in situ, we investigated the relationship between sleep and working memory in a real-world environment. The results show that in a real-world environment, working memory increases as the average HR and average HRV during sleep increase. When stress levels are low, average HRV increases while average HR decreases. Therefore, when investigating the impact of stress on working memory, it is essential to consider these respective changes in future work. The fit of the model constructed in this study was surprisingly high, considering the experimental conditions. This suggests that physiological data collected by sleep-tracking devices can obtain information on the relationship between sleep and working memory even in real-world environments. Future plans include re-examining the methods for evaluating working memory and identifying factors other than sleep data, such as room temperature, that affect working memory to improve the goodness of fit.

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REFERENCES

- [1] Hirotugu Akaike. 1987. Factor analysis and AIC. *Psychometrika* 52 (1987), 317–332.
- [2] Michael Olusegun Akinwande, Hussaini Garba Dikko, Agboola Samson, et al. 2015. Variance inflation factor: as a condition for the inclusion of suppressor variable (s) in regression analysis. *Open journal of statistics* 5, 07 (2015), 754.
- [3] Alan Baddeley. 1992. Working memory. *Science* 255, 5044 (1992), 556–559.
- [4] Rachael N Blasiman and Christopher A Was. 2018. Why is working memory performance unstable? A review of 21 factors. *Europe's journal of psychology* 14, 1 (2018), 188.

- [5] Rui Cao, Iman Azimi, Fatemeh Sarhaddi, Hannakaisa Niela-Vilen, Anna Axelin, Pasi Liljeberg, and Amir M Rahmani. 2022. Accuracy assessment of oura ring nocturnal heart rate and heart rate variability in comparison with electrocardiography in time and frequency domains: comprehensive analysis. *Journal of Medical Internet Research* 24, 1 (2022), e27487.
- [6] Rossana Castaldo, Paolo Melillo, Umberto Bracale, M Caserta, Maria Triassi, and Leandro Pecchia. 2015. Acute mental stress assessment via short term HRV analysis in healthy adults: A systematic review with meta-analysis. *Biomedical Signal Processing and Control* 18 (2015), 370–377.
- [7] Andrew RA Conway, Michael J Kane, and Randall W Engle. 2003. Working memory capacity and its relation to general intelligence. *Trends in cognitive sciences* 7, 12 (2003), 547–552.
- [8] Giuseppe Curcio, Michele Ferrara, and Luigi De Gennaro. 2006. Sleep loss, learning capacity and academic performance. *Sleep medicine reviews* 10, 5 (2006), 323–337.
- [9] Patrick D Gajewski, Eva Hanisch, Michael Falkenstein, Sven Thönes, and Edmund Wascher. 2018. What does the n-back task measure as we get older? Relations between working-memory measures and other cognitive functions across the lifespan. *Frontiers in psychology* 9 (2018), 2208.
- [10] Cameron B Jeter, Saamil S Patel, and Anne B Sereno. 2011. Novel n-back spatial working memory task using eye movement response. *Behavior research methods* 43 (2011), 879–887.
- [11] Kazuko Kato, Kunihiro Iwamoto, Naoko Kawano, Yukihiro Noda, Norio Ozaki, and Akiko Noda. 2018. Differential effects of physical activity and sleep duration on cognitive function in young adults. *Journal of sport and health science* 7, 2 (2018), 227–236.
- [12] Hye-Geum Kim, Eun-Jin Cheon, Dai-Seg Bai, Young Hwan Lee, and Bon-Hoon Koo. 2018. Stress and heart rate variability: a meta-analysis and review of the literature. *Psychiatry investigation* 15, 3 (2018), 235.
- [13] Wayne K Kirchner. 1958. Age differences in short-term retention of rapidly changing information. *Journal of experimental psychology* 55, 4 (1958), 352.
- [14] Clemens Kirschbaum, Karl-Martin Pirke, and Dirk H Hellhammer. 1993. The 'Trier Social Stress Test'—a tool for investigating psychobiological stress responses in a laboratory setting. *Neuropsychobiology* 28, 1-2 (1993), 76–81.
- [15] Kenichi Kuriyama, Kazuo Mishima, Hiroyuki Suzuki, Sayaka Aritake, and Makoto Uchiyama. 2008. Sleep accelerates the improvement in working memory performance. *Journal of Neuroscience* 28, 40 (2008), 10145–10150.
- [16] Milad Asgari Mehrabadi, Iman Azimi, Fatemeh Sarhaddi, Anna Axelin, Hannakaisa Niela-Vilén, Saana Myllyntausta, Sari Stenholm, Nikil Dutt, Pasi Liljeberg, Amir M Rahmani, et al. 2020. Sleep tracking of a commercially available smart ring and smartwatch against medical-grade actigraphy in everyday settings: instrument validation study. *JMIR mHealth and uHealth* 8, 11 (2020), e20465.
- [17] Oliver Monfredi, Alexey E Lyashkov, Anne Berit Johnsen, Shin Inada, Heiko Schneider, Ruoxi Wang, Mahesh Nirmalan, Ulrik Wisloff, Victor A Maltsev, Edward G Lakatta, Henggui Zhang, and Mark R Boyett. 2014. Biophysical Characterization of the Underappreciated and Important Relationship Between Heart Rate Variability and Heart Rate. *Hypertension* 64 (2014), 1334–1343. Issue 6. <https://doi.org/10.1161/HYPERTENSIONAHA.114.03782>
- [18] Shinichi Nakagawa and Holger Schielzeth. 2013. A general and simple method for obtaining R² from generalized linear mixed-effects models. *Methods in ecology and evolution* 4, 2 (2013), 133–142.
- [19] Adrian M. Owen, Kathryn M. McMillan, Angela R. Laird, and Ed Bullmore. 2005. N-back working memory paradigm: A meta-analysis of normative functional neuroimaging studies. *Human Brain Mapping* 25, 1 (2005), 46–59. <https://doi.org/10.1002/hbm.20131> arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1002/hbm.20131>
- [20] Kimberly P Raghubar, Marcia A Barnes, and Steven A Hecht. 2010. Working memory and mathematics: A review of developmental, individual difference, and cognitive approaches. *Learning and individual differences* 20, 2 (2010), 110–122.
- [21] Heike Schmidt, Jigar Jogia, Kristina Fast, Tessa Christodoulou, Morgan Haldane, Veena Kumari, and Sophia Frangou. 2009. No gender differences in brain activation during the N-back task: An fMRI study in healthy individuals. *Human brain mapping* 30, 11 (2009), 3609–3615.
- [22] Joachim Taelman, Steven Vandeput, Arthur Spaepen, and Sabine Van Huffel. 2009. Influence of mental stress on heart rate and heart rate variability. In *4th European Conference of the International Federation for Medical and Biological Engineering: ECIFMBE 2008 23–27 November 2008 Antwerp, Belgium*. Springer, 1366–1369.
- [23] Benjamin Tag, Tilman Dingler, Andrew W. Vargo, and Vassilis Kostakos. 2020. Inferring Circadian Rhythms of Cognitive Performance in Everyday Life. *IEEE Pervasive Computing* 19, 3 (July 2020), 14–23. <https://doi.org/10.1109/MPRV.2020.2994914> Conference Name: IEEE Pervasive Computing.
- [24] Andrew Vargo, Masayuki Fujimoto, and Koichi Kise. 2023. Sleep Quality Tracking and Alignment with a Simple In-the-Wild Task. In *Adjunct Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing and the 2022 ACM International Symposium on Wearable Computers (UbiComp/ISWC '22 Adjunct)*. Association for Computing Machinery, New York, NY, USA, 128–129. <https://doi.org/10.1145/3544793.3560359>
- [25] Oliver Wilhelm, Andrea Hildebrandt, and Klaus Oberauer. 2013. What is working memory capacity, and how can we measure it? *Frontiers in psychology* 4 (2013), 433.